

Maximal abelian *-subalgebras

`MathlibAnnex.Analysis.CStarAlgebra.IsMaximalAbelian`

def

Defines maximality by inclusion among commutative unital *-subalgebras.

Statement

Let A be a unital complex C^* -algebra and let $D \subseteq A$ be a unital complex *-subalgebra. The predicate below says that D is commutative and that no strictly larger commutative unital *-subalgebra contains it.

Definition

Explicitly, D is maximal abelian precisely when both conditions hold:

$$xy = yx \quad (x, y \in D),$$

and, for every commutative unital *-subalgebra E ,

$$D \subseteq E \implies E \subseteq D.$$

In the second condition the two inclusions force $E = D$. The comparison is with all such subalgebras, not only with norm-closed ones.

Assumptions

D contains the unit and is closed under addition, complex scalar multiplication, multiplication and adjoints. Norm closedness is not an assumption in this definition.

Conclusion

This is a property of the given subalgebra D , rather than a construction of a new subalgebra.

Norm closedness of a maximal abelian subalgebra is a later consequence, not an extra conjunct of this predicate.

Main citations

- Exact definition and context

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- A maximal abelian unital star subalgebra, expressed without installing a
global commutative-ring instance on its subtype. -/
def IsMaximalAbelian (D : StarSubalgebra ℂ A) : Prop :=
  IsMulCommutative D ∧
  ∀ E : StarSubalgebra ℂ A, IsMulCommutative E → D ≤ E → E ≤ D
```

In the source	Mathematical meaning
D : StarSubalgebra ℂ A	The specified unital complex *-subalgebra D of A .
: Prop	The output is a condition that may or may not hold for D .
IsMulCommutative D	For any two elements x, y of D , their products satisfy $xy = yx$.
∀ E : StarSubalgebra ℂ A	Quantify over every unital complex *-subalgebra E of the same A .
IsMulCommutative E → D ≤ E → E ≤ D	If E is commutative and contains D , then every element of E already lies in D ; hence $E = D$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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